homework-03

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Set up

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggeffects)

# store personal data
personal_data <- read_csv(here( "data" , "personaldataproject.csv"))

# store kelp data
temp_kelp <- read_csv(here("data", "temp-kelp.csv"))
```

Problems

Problem 1. Giant Kelp Fronds

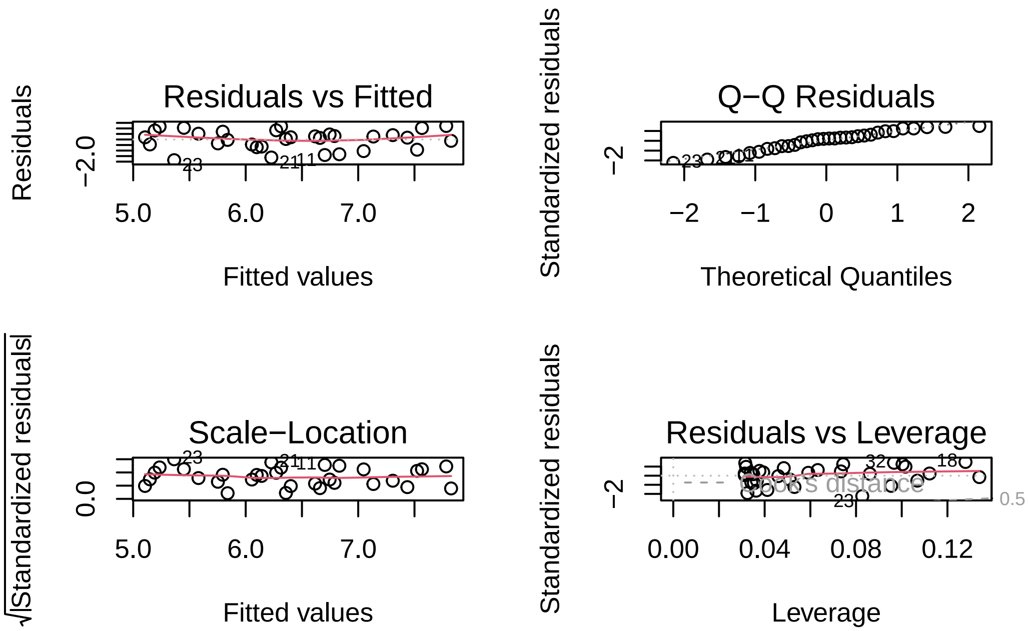
a. An Appropriate Test

Pearson and Spearman correlations could be used to determine strength of the relationship between temperature and giant kelp frond elongation rate assuming observations are independent. Pearson's correlation also assumes a linear relationship exists and it is a parametric test meaning it assumes variables are normally distributed. Spearman's correlation assumes a monotonic relationship and it is a nonparametric test meaning normality is not an assumption.

b. Visualizing Data

```
# create object storing linear model data to perform diagnostic tests
temp_kelp_model <- lm(kelp_elong~temp_c,
   data = temp_kelp)

# diagnostic plots
par(mfrow = c(2,2))
plot(temp_kelp_model)
```




```
# display summary for model coefficients
summary(temp_kelp_model)
```

```
Call:
lm(formula = kelp_elong ~ temp_c, data = temp_kelp)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8643 -0.5017  0.1790  0.5659  1.2177 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473, Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

```
#creating new object called temp_kelp_preds
temp_kelp_preds <- ggpredict(
  temp_kelp_model,  # model object
  terms = "temp_c") # predictor variable

#display predictions
temp_kelp_preds
```

```
# Predicted values of kelp_elong

temp_c | Predicted |     95% CI
-------------------------------
 14.50 |      7.83 | 7.18, 8.47
 15.50 |      7.39 | 6.89, 7.90
 16.00 |      7.18 | 6.74, 7.62
 17.00 |      6.75 | 6.40, 7.09
 18.00 |      6.31 | 6.00, 6.63
 18.50 |      6.10 | 5.77, 6.43
 19.50 |      5.67 | 5.24, 6.09
 21.00 |      5.02 | 4.40, 5.64
```

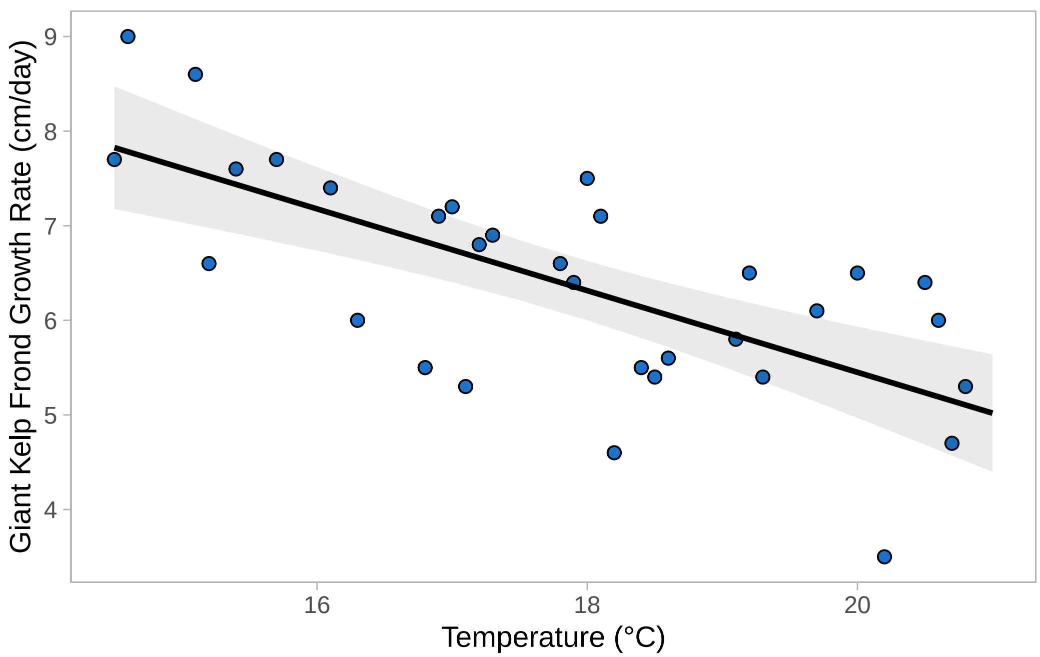
```
# plot raw data points to observe potential relationship

# base layer ggplot
ggplot(data = temp_kelp,
       # assign data to axes
       aes(x = temp_c,
           y = kelp_elong))+
# first layer: add points
  geom_point(size = 2,
             stroke = 0.5,
             fill = "dodgerblue3",
             shape = 21)+

# second layer: ribbon representing 95% CI
# using predictions data frame
geom_ribbon(data = temp_kelp_preds,
            aes(x = x,
                y = predicted,
                ymin = conf.low,
                ymax = conf.high),
```

```
              alpha = 0.1) +
# third layer: line representing model predictions
# using predictions data frame
geom_line(data = temp_kelp_preds,
          aes (x = x,
               y = predicted),
          linewidth = 1) +

# relabel axis with units
  labs(x = "Temperature (°C)",
       y = "Giant Kelp Frond Growth Rate (cm/day)")+
#change theme
    theme_light()+
    theme(panel.grid = element_blank())
```




```
# data appears fairly linear, proceed with diagnostic tests
```

c. Check Assumptions

Assumptions of linear regression:

- Linear relationship expected: A linear relationship is logical. I would expect as temperature increases kelp frond growth rate would decrease since they thrive in cool water.
- Independent errors: We are not given much information about data collection, so I am going to assume there is no correlation between errors
- Homoscedasticity: Looking at diagnostic plots in 1b, it appears the homoscedasticity of error is present since the points are roughly evenly and randomly distributed.
- Normally Distributed Error: Looking at diagnostic plots in 1b, the QQ plot suggests normal distribution since the data follows the line closely.

Assumption of Pearson's correlation

- Linear relationship between variables: looking at the geom point graph it appears linear. By looking at the R^2 value calculated in 1b we can say 47% of the variability can be accounted for by the predictor variable (temperature) in the linear model summary. We can further assess the strength of the relationship with a correlation test.
- Variables are continuous: Temperature and length are both continuous variables (they can be measured to decimal places or fractions)
- Variables are normally distributed (discussed above)
- Independent observations (discussed above)

Run a correlation test

```r
# Run Pearson Correlation test
cor.test(temp_kelp$kelp_elong, # predicted/y value
         temp_kelp$temp_c,     # predictor/x value
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  temp_kelp$kelp_elong and temp_kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d.

e.

f.

Problem 2. Personal Data

a.

b.

Problem 3. Affective Visualization

a.

b.

c.

d.

e.

Problem 4. Statistical Critique

a.

b.

c.

d.